



Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S. Manufacturing

Final Presentation

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Executive Summary

1

Objective: Do reported prevention actions actually lead to lower toxic releases the next year — and can we predict which facilities will fall short?

2

Business Problem: EHS teams need a practical way to know which [Environment Protection Agency \(EPA\)'s Toxic Release Inventory \(TRI\)](#) records deserve review first.

3

Data: EPA TRI Basic Plus Form R records from 2020 to 2024, focused on U.S. manufacturing.

4

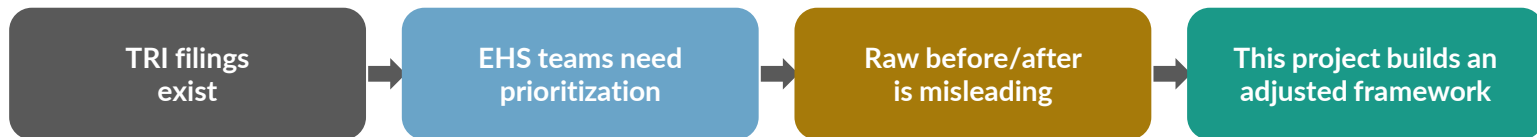
Method: Statistical tests plus an interpretable elastic-net logistic regression model.

5

Main Takeaway: Reported actions are useful but modest signals; the model should support human review, not replace experts.



Gap Analysis



What Prior Research Established

- Public disclosure changes behavior and moves markets (Hamilton, 1995; Konar & Cohen, 1997).
- Some of the apparent national decline reflects reporting change, not real abatement (Gamper-Rabindran, 2006).
- Prevention is not one undifferentiated thing; the profitable part is preventing waste, not treating it (King & Lenox, 2002; Cagno et al., 2005).

The Gap This Project Fills

- Prior work asks broader questions (disclosure effects, long-run emissions); it does not give EHS teams a ready-to-use screening tool.
- This project is narrower and current: 2020 to 2024 facility-chemical-year records only.
- It defines one clear, testable outcome: next-year release no more than 90% of current-year release.
- It is an association and screening study, not a causal claim.



Research Questions

1 of 2

RQ1: *Do reduction rates differ by action year and manufacturing subsector?*

Hypotheses

H0: Reduction is independent of action year and of subsector. **Ha:** Reduction rates differ across years and across subsectors.

Method & Why

Chi-square test, Cramer's V: the predictor is categorical and the sample is large; Cramer's V checks whether a significant effect is actually large enough to act on

Result

Both nulls rejected. Rates rise 27.5% to 31.1% to 35.9% across year transitions ($p < .001$, $V = 0.073$); rates span about 15 points across subsectors, 25.9% to 41.3% ($p < .001$, $V = 0.048$). Effects are real but small; year and subsector must be controlled downstream.

RQ2: *Is reporting any action associated with next-year reduction, after adjustment?*

Hypotheses

H0: The adjusted odds ratio for reporting any action equals 1. **Ha:** The adjusted odds ratio differs from 1.

Method & Why

Facility-clustered logistic regression: outcome is binary so coefficients convert into odds ratios; clustering is required because one facility contributes multiple records, so observations are not independent

Result

Null rejected. Adjusted odds ratio = 1.14, 95% CI 1.06 to 1.23, $p < .001$. A reported action raises the odds of a meaningful reduction by about 14%, net of confounders. Model diagnostics are clean (max VIF about 2).



Research Questions

2 of 2

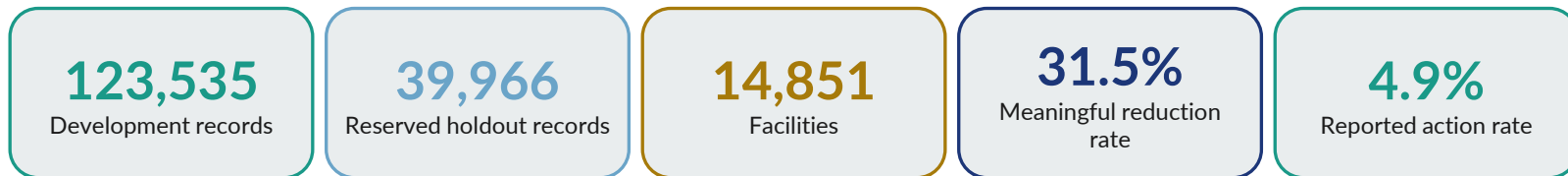
RQ3: *Do reduction rates differ between each action type and no-action records?*

Hypotheses	H0: For each action type, the reduction rate equals the no-action rate. Ha: At least one action type differs from the no-action group after Holm correction.
Method & Why	Holm-corrected pairwise comparisons: five action types are each compared against one shared no-action group; Holm's correction controls family-wise error with more power than Bonferroni
Result	Two of five nulls rejected. Material modification: 39.0% rate, +7.6 points, Holm $p < .001$. Operating practice/training: 35.5% rate, +4.2 points, Holm $p < .001$. The other three action types did not survive correction.

RQ4: *Can an interpretable model predict non-reduction better than chance and a baseline?*

Hypotheses	H0: The model's holdout AUC equals 0.5 (chance) and does not exceed the baseline. Ha: The model exceeds chance and the baseline on the sealed holdout.
Method & Why	Elastic-net penalized logistic regression: stays interpretable, handles correlated predictors such as 15 subsectors and 5 action types, and avoids overfitting; evaluated once on a sealed 2023-2024 holdout
Result	Null rejected. Holdout AUC = 0.523, 95% CI 0.515 to 0.532, vs baseline 0.502. The model beats chance and the baseline, but the margin is small, useful for triage, not automation.

Data Description and EDA

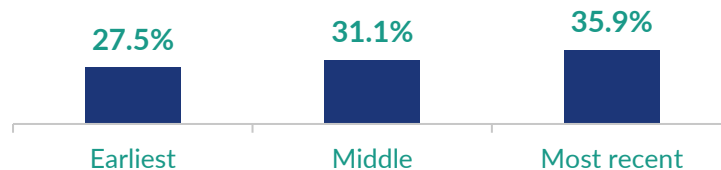


Source: Environment Protection Agency (EPA's) Toxic Release Inventory Basic Plus Form R filings, 2020 to 2024, U.S. manufacturing only.

Outcome Balance (Development Set)

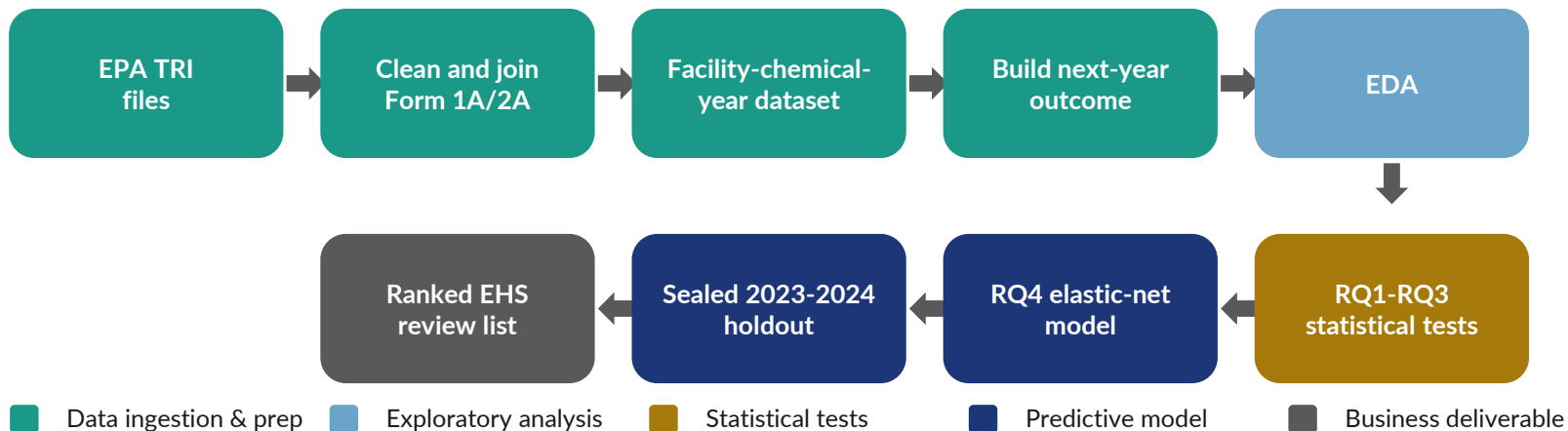


Reduction Rate by Action-Year Transition



Takeaway: the outcome is learnable, but reported actions are rare, so the analysis needs careful, adjusted inference.

Architecture diagram/Workflow



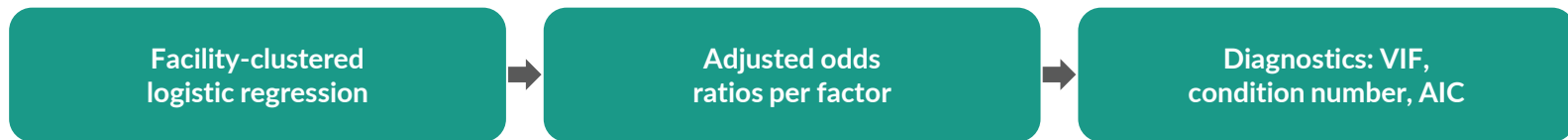
Note: the holdout year was used once, only after the model design was locked.



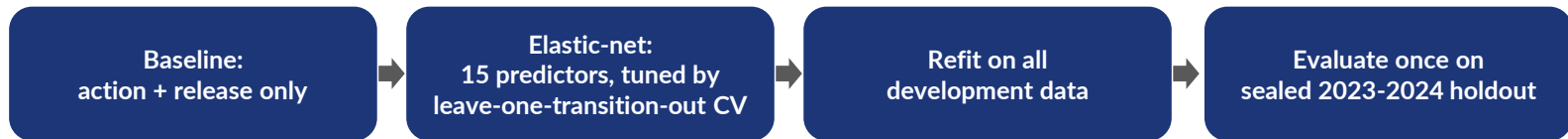
Model Building

1 of 2

Explanatory Path: (why)



Predictive Path: (how well)



Why two paths: RQ2 answers whether reported actions matter (an adjusted association, for explanation). RQ4 answers how well those factors can predict next year's outcome in advance (for a screening tool). The deck's Results reflect both.



Model Building

2 of 2

Choice	Why it was chosen	What it protects against
Logistic regression	Outcome is binary; coefficients convert into explainable odds ratios.	<i>Overly complex models that a business reader cannot interpret.</i>
Facility-clustered standard errors	Facilities appear multiple times in the data.	<i>Understated standard errors from treating repeated records as independent.</i>
Holm correction	Five action-type comparisons share one reference group.	<i>False positives from testing five hypotheses at once.</i>
Elastic-net logistic regression	Interpretable; handles 15 subsector and 5 action-type predictors that correlate with each other.	<i>Overfitting to development-set noise.</i>
Out-of-time holdout (2023 to 2024)	Mirrors real business use: train on the past, score the future.	<i>Optimistic results from testing on data the model has already seen.</i>

Diagnostics confirm the model is sound: maximum variance inflation factor about 2 (low multicollinearity); condition number 81 (stable); Akaike Information Criterion favors the quadratic production-ratio term over a linear one (150,795 vs 150,817).



Results

1 of 2

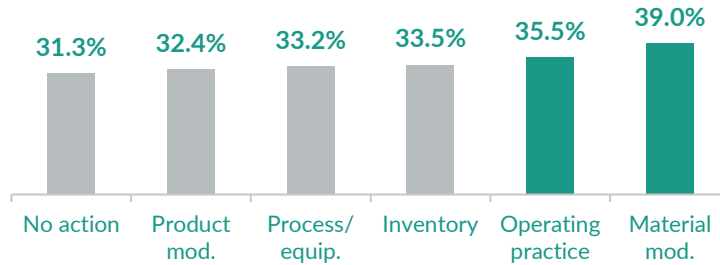
Metric	Result	What it means
Any reported action (odds ratio)	1.14	Modestly higher odds of reduction, after adjusting for other factors.
Material modification	39.0% rate (+7.6 pts vs no action)	Strongest action type.
Operating practice / training	35.5% rate (+4.2 pts vs no action)	Second-strongest action type.
Elastic-net model AUC	0.523 vs baseline 0.502	Slightly better than chance and the simple baseline.
Brier score (calibration)	0.218	Reasonably well-calibrated probabilities.

The signal is real but modest. The model is useful for triage, not automation.

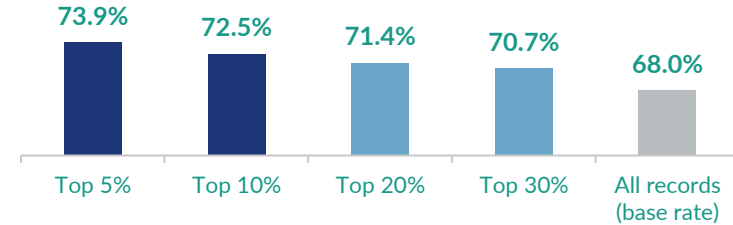
Results

2 of 2

Reduction Rate by Action Type



Screening Lift: Non-Reduction Rate by Review Depth



- **Left:** only material modification and operating practice/training (teal) separate from the other action types after correction; the rest are statistically indistinguishable from no action (grey).
- **Right:** the ranked list concentrates hard-to-improve records near the top (73.9% vs a 68.0% base rate); the lift fades with depth, so review value is highest at the very top of the list.



Implementation and User Benefit

1

Triage With the Model

Top 5% of ranked records show a 73.9% non-reduction rate vs a 68.0% base rate. Hold review capacity fixed and work the list top-down each cycle.

2

Focus Prevention Where It Works

Only material modification and operating practice/training show a confirmed lift after correction. Prioritize funding and pilots there; share results across sites.

3

Weight by Pounds at Risk

Every scored record also carries its release quantity. Rank by risk probability multiplied by release volume, so effort follows impact, not just likelihood.

4

Benchmark Peers, Not Zero

Reduction rates span about 15 points across subsectors and rise over time. Compare each facility to its peer group and to trend, not to last year alone.

Human experts stay in control. *the model tells the team where to look first.*

Green initiative: *connect the ranked worklist to green investment planning, incentive tracking, and sustainability reporting.*



Conclusion

Key Findings

- Reported pollution-prevention actions contain useful information.
- Strongest signals: material modification and operating practice/training.
- Year and industry matter, so raw before-and-after comparisons are not enough.
- Best use: a transparent decision-support tool for EHS prioritization.

Limitations

- Observational study; results are associations, not proof of cause and effect.
- Predictive lift is modest.
- Reported actions are rare, under 5 percent of records, which limits precision.

Future Work

- Add production, cost, plant process, and sustainability investment data where available.



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